V2 Answers

1 (a) This is preferable for increased speed/high frequency performance [2] This is because NPN transistors generally have higher FTEs than PNP transistors [1]. This is due to mobility of electrons being approximately 3x that of holes [2].

(b) Same as V1.

(d) Different numerical answer (method same).

2 (b, e) Different numerical answer (method same).

3 (b) Different numerical answer (method same).

3 (e) (i) Increase r0 of M9 can be done by cascading.

(ii) Same as V1